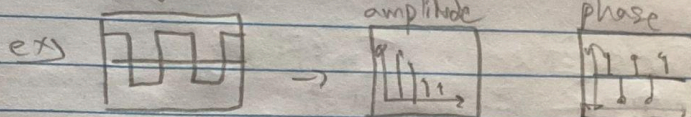


Image Processing II

Fourier said that any periodic function can be rewritten as weighted sum of infinite sinusoids at different frequencies

$$\text{Sinusoid: } f(x) = A \sin(2\pi u x + \phi)$$

A = amplitude, u = Frequency, ϕ = shift / phase



no loss of information! $f(x) \xrightarrow{\text{FT}} F(u)$
 $F(u)$ is amplitude at freq. u
 $f(x) \xleftarrow{\text{IFT}} F(u)$

$$\text{Fourier Transform: } F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi u x} dx$$

x = space

u = freq

$e^{i\theta} = \cos\theta + i\sin\theta$

$i = \sqrt{-1}$

$$\text{Inverse FT: } f(x) = \int_{-\infty}^{\infty} F(u) e^{i2\pi u x} dx$$

Why does $e^{i\theta} = \cos\theta + i\sin\theta$

- expand w/ Taylor Series

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \dots$$

$$= (1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots) + i(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots)$$

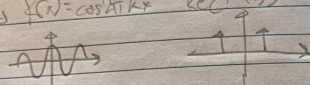
$$= \cos\theta + i\sin\theta$$

$F(u)$ holds amp and phase w/ freq u

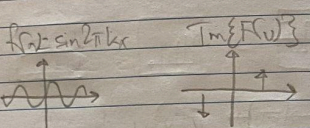
$$F(u) = \text{Re}\{F(u)\} + i \text{Im}\{F(u)\}$$

Amplitude: $A(u) = \sqrt{\text{Re}\{F(u)\}^2 + \text{Im}\{F(u)\}^2}$
 Phase: $\phi(u) = \arctan\left(\frac{\text{Im}\{F(u)\}}{\text{Re}\{F(u)\}}\right)$

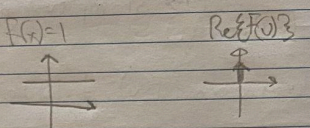
cs) $f(x) = \cos 2\pi kx$ $\text{Re}\{F(u)\}$



$f(x) = \sin 2\pi kx$ $\text{Im}\{F(u)\}$



$f(x) = 1$ $\text{Re}\{F(u)\}$



$f(x) = e^{-ax^2}$ $F(u) = \sqrt{\frac{\pi}{a}} e^{-\frac{\pi^2 u^2}{a}}$

shows inverse scaling

Properties of FT

- Linearity: $\alpha f_1(x) + \beta f_2(x) \rightarrow \alpha F(u) + \beta F(u)$
- Scaling: $f(ax) \rightarrow \frac{1}{|a|} F\left(\frac{u}{a}\right)$
- Shifting: $f(x-a) \rightarrow e^{-2\pi i u a} F(u)$
- Differentiation: $\frac{d}{dx} f(x) \rightarrow (i2\pi u) F(u)$

Convolution Theorem

$$g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(x') h(x-x') dx'$$

$$\downarrow \text{FT}$$

$$G(u) = \int_{-\infty}^{\infty} g(x) e^{-i2\pi u x} dx$$

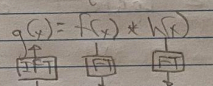
$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x') h(x-x') e^{-i2\pi u x} dx' dx$$

$$= \int_{-\infty}^{\infty} f(x') e^{-i2\pi u x'} dx' \int_{-\infty}^{\infty} h(x-x') e^{-i2\pi u (x-x')} dx$$

$$= F(u) H(u)$$

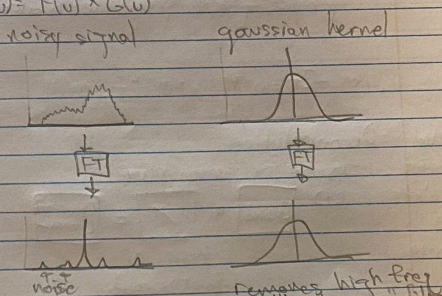
Thus convolution in space domain $\leftrightarrow$ multiplication in frequency domain

$g(x) = f(x) * h(x)$



$G(u) = F(u) \times H(u)$

cs) noisy signal gaussian kernel



removes high freq $\rightarrow$ "high pass" filter

Extending to 2D

-FT

$$F(u,v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) e^{-i2\pi(ux+vy)} dx dy$$

-IFT

$$f(x,y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u,v) e^{i2\pi(ux+vy)} du dv$$

Discretely...

$$F(p,q) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m,n] e^{-i2\pi pm/M} e^{-i2\pi qn/N}$$

$p=0 \dots M-1, q=0 \dots N-1$

$$f[m,n] = \frac{1}{MN} \sum_{p=0}^{M-1} \sum_{q=0}^{N-1} F(p,q) e^{i2\pi pm/M} e^{i2\pi qn/N}$$

$m=0 \dots M-1, n=0 \dots N-1$

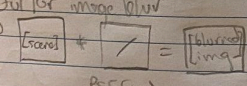
p and q are frequencies along m and n respectively

$\cos -2\pi pm/M, \sin -2\pi pm/M$

IMU (inertial measurement unit) in phone can help determine PSF

Deconvolution

-useful for image blur

cs) 

PSF (Point Spread Function)

Let F' be recovered scene

$$F(x,y) * h(x,y) = g(x,y)$$

$$F(u,v) H(u,v) = G(u,v)$$

$$F(u,v) = \frac{G(u,v)}{H(u,v)} \rightarrow \text{[FT]} \rightarrow F'$$

But noise complicates this

- added to img after $f(x,y) * g(x,y) + n(x,y) = g(x,y)$
- we can't afford to ignore it
- if $H(u,v) = 0$, F' is not recoverable (div. by 0)
- motion blur filter $H(u,v)$ is low pass filter
- For high freqs (u,v):
 - noise $N(u,v)$ in $g(u,v)$ is high
 - filter $H(u,v) \approx 0$
- Thus noise is amplified

Wiener Deconvolution

$$F'(u,v) = \frac{G(u,v)}{H(u,v)} \left(\frac{1}{1 + \frac{\text{NSR}(u,v)}{|H(u,v)|^2}} \right)$$

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NSR = noise-to-signal ratio $\rightarrow$ we don't know! or find

$$\text{NSR}(u,v) = \frac{\text{power of noise at } (u,v)}{\text{power of signal (scene) at } (u,v)} = \frac{N(u,v)^2}{|F(u,v)|^2}$$

NSR often set to constant λ (eg. $\lambda = 0.002$)